

ATHUL DEV

AI Engineer | LLM Applications & Production System Engineer

Kochi, Kerala, India | +91 7034306102 | athuldev743@gmail.com | port-folio-phpa.vercel.app

SUMMARY

AI Engineer specializing in production LLM applications, agentic workflows, RAG architectures, and multi-provider fallback infrastructure. Builds end-to-end tool-calling agents, vector retrieval pipelines, and scalable FastAPI backends across MongoDB and PostgreSQL. Experienced in hallucination guardrails, semantic search optimization, and low-latency API integration to maintain high availability under provider outages and rate limits.

SKILLS

- **AI & LLM Engineering:** LLM (Large Language Model) Agents, Tool-Calling Architectures, RAG (Retrieval-Augmented Generation), Semantic Search, Vector DBs (ChromaDB), SentenceTransformers, Multi-Provider Fallback, Prompt Cadence Logic
- **Programming & Core:** Python (Asyncio), SQL, Data Structures, Linux, Git
- **Backend APIs:** FastAPI, RESTful APIs (Application Programming Interfaces), WebSockets, Event-Driven Architecture, Microservices, Connection Pooling
- **Databases & Storage:** PostgreSQL (Neon, Aiven), MongoDB (Motor, Beanie), SQLite, Redis, asyncpg
- **Automation & Cloud:** FastAPI Cloud, APScheduler, QStash, Webhooks, Railway, Cloudflare Tunnels, Cloudinary CDN, Docker
- **Security & Reliability:** HMAC signature verification, JWT (JSON Web Tokens) / OAuth 2.0, idempotent handling, load testing (Locust), query profiling (EXPLAIN ANALYZE)

AI ENGINEERING PROJECTS

DOOM — Personal AI Assistant

Nov 2025 | doom-1a9d9743.fastapicloud.dev

- Architected a ReAct-style agent orchestrating **35 specialized tools** for job discovery/apply, email, WhatsApp, contacts, RAG, and PC automation under strict tool-calling protocols and hallucination guardrails preventing false claims.
- Achieved zero-downtime AI availability via a **4-provider, 14-model fallback chain** (Gemini → OpenAI → Groq → OpenRouter), handling rate limits and provider outage events cleanly.
- Built a ChromaDB and SentenceTransformers RAG pipeline with overlap-aware chunking and re-ingestion deduplication; automated job-discovery and lead-enrichment workflows integrating Naukri, LinkedIn, Snov.io, and Gmail API.

Neolix — B2B Sales Outreach Automation

Feb 2026 | neolix-sage.vercel.app

- Built a 9-day, 3-channel (Email/WhatsApp/SMS) outreach cadence engine with context-aware AI copywriting using Groq Llama 3.3 70B/8B (routed dynamically by cost-latency tradeoff) and Gmail API (OAuth 2.0) integration.
- Reduced median API latency 66% (2700ms → 920ms) by diagnosing database connection limits via Locust load testing and implementing asyncpg connection pooling; verified sub-millisecond GIN-indexed full-text search over a 1M+ profile dataset.
- Fixed a cross-database architecture bug where a PostgreSQL query referenced a non-existent dedup table, replacing it with a MongoDB-backed lookup aligned with system design.

PROFESSIONAL EXPERIENCE

Web Developer & AI Engineer — OquiliX

Kochi, India | July 2025 – Present

- Improved AI query relevance through vector-based semantic search pipelines, reducing manual data-retrieval overhead.
- Architected production-grade FastAPI REST APIs for e-commerce inventory and order processing across concurrent user sessions.
- Designed asynchronous database architectures supporting non-blocking read/write operations and improved reliability under high-traffic loads.

SYSTEMS WEB PROJECTS

Ekabhumi — E-Commerce Platform

July 2025 | www.ekabhumi.in

- Designed a three-layer payment verification system (client HMAC-SHA256, server Razorpay cross-check, separately-keyed webhooks) with idempotent processing to block duplicate fulfillment.
- Built Google OAuth 2.0 authentication issuing self-signed JWTs with server-side role assignment and silent refresh.

EDUCATION

Bachelor of Computer Applications (BCA) — University of Calicut

2022 – 2025